

Kit de survie d'analyse

Formules générales théoriques

(1)  $\boxed{\int_a^x f'(t) \, dt = f(x)}$  (Théorème fondamental de l'analyse)

(2)  $\int u'v = uv - \int v'u,$  (Formule d'intégration par parties)

(3)  $\int_{u(a)}^{u(b)} f(x) \, dx = \int_a^b f(u(t))u'(t) \, dt$  (Changement de variable)

Primitives de bases

(4)  $\int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1)$

(5)  $\int \frac{1}{ax+b} dx = \frac{1}{a} \ln |ax+b| \quad (a \in \mathbb{R}^*, b \in \mathbb{R})$

(6)  $\int e^{\lambda x} dx = \frac{1}{\lambda} e^{\lambda x} \quad (\lambda \neq 0)$

(7)  $\int \sin(x) dx = -\cos(x) \quad \int \cos(x) dx = \sin(x)$

(8)  $\int \tan(x) dx = -\ln |\cos(x)|$

(9)  $\int \frac{1}{\tan x} dx = \ln |\sin x|$

(10)  $\int \frac{1}{\cos^2 x} dx = \tan(x) \quad \int \frac{1}{\sin^2 x} dx = -\frac{1}{\tan x}$

(11)  $\int \frac{1}{\sin(x)} dx = \ln \left| \tan\left(\frac{x}{2}\right) \right|$

(12)  $\int \frac{1}{\cos(x)} dx = \ln \left| \tan\left(\frac{x}{2} + \frac{\pi}{4}\right) \right|$

(13)  $\int \operatorname{ch}(x) dx = \operatorname{sh}(x) \quad \int \operatorname{sh}(x) dx = \operatorname{ch}(x)$

(14)  $\int \operatorname{th}(x) dx = \ln(\operatorname{ch}(x))$

(15)  $\int \frac{1}{\operatorname{th}(x)} dx = \ln|\operatorname{sh}(x)|$

(16)  $\int \frac{1}{\operatorname{ch}(x)} dx = 2 \cdot \arctan(e^x)$

(17)  $\int \frac{1}{\operatorname{sh}(x)} dx = \ln \left| \operatorname{th}\left(\frac{x}{2}\right) \right|$

(18)  $\int \frac{1}{\operatorname{ch}^2(x)} dx = \operatorname{th}(x) \quad \int \frac{1}{\operatorname{sh}^2(x)} dx = -\frac{1}{\operatorname{th}(x)}$

(19)  $\int \frac{1}{\sqrt{x^2+1}} dx = \ln(x + \sqrt{x^2+1})$   
 $= \operatorname{argsh}(x)$

(20)  $\int \frac{1}{\sqrt{x^2-1}} dx = \ln(x + \sqrt{x^2-1})$   
 $= \operatorname{argch}(x)$

(21)  $\int \frac{1}{x^2-1} dx = \ln\left(\frac{x-1}{x+1}\right) = \operatorname{argth}(x)$

(22)  $\int \frac{1}{x^2+1} dx = \arctan(x)$

(23)  $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x)$  ou  $-\arccos(x)$

(24)  $\int \sqrt{x^2-1} dx = \frac{x}{2} \sqrt{x^2-1} + \frac{1}{2} \ln(x + \sqrt{x^2-1})$

(25)  $\int \sqrt{x^2+1} dx = \frac{x}{2} \sqrt{x^2+1} + \frac{1}{2} \ln(x + \sqrt{x^2+1})$

(26)  $\int \sqrt{1-x^2} dx = \frac{x}{2} \sqrt{1-x^2} + \frac{1}{x} \arcsin(x)$

Développement limités usuels

(1)  $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots + x^n + o(x^n)$

(2)  $\ln(1-x) = -x - \frac{x^2}{2} - \dots - \frac{x^n}{n} + o(x^n)$

(3)  $\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots + (-1)^n x^n + o(x^n)$

(4)  $\ln(1+x) = x - \frac{x^2}{2} + \dots + (-1)^{n+1} \frac{x^n}{n} + o(x^n)$

(5)  $\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots + (-1)^n x^{2n} + o(x^n)$

(6)  $\operatorname{argth}(x) = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots + \frac{x^{2n+1}}{(2n+1)} + o(x^{2n+2})$

(7)  $\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)} + o(x^{2n+2})$

(8)  $(1+x)^\alpha = 1 + \alpha x + \dots + \frac{\alpha(\alpha-1) \dots (\alpha-n+1)}{n!} x^n + o(x^n)$

(9)  $e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + o(x^n)$

(10)  $\operatorname{ch}(x) = \frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + \frac{x^{2n}}{(2n)!} + o(x^{2n+1})$

(11)  $\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!} + o(x^{2n+1})$

Relations remarquables

(12) 
$$\operatorname{sh}(x) = \frac{e^x - e^{-x}}{2} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$$

(13) 
$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$$

(14) 
$$\tan x = x + \frac{2}{3}x^3 + \frac{17}{15}x^5 + \frac{17}{315}x^7 + \cdots + \frac{(-4)^n(1-4^n)B_{2n}}{(2n)!}x^{2n-1} + o(x^{2n})$$

(15) 
$$\operatorname{th}(x) = x - \frac{x^3}{3} + \frac{2}{15}x^5 - \frac{17}{315}x^7 + \cdots + \frac{4^n(1-4^n)B_{2n}}{(2n)!}x^{2n-1} + o(x^{2n})$$

(16) 
$$\arccos(x) = \frac{\pi}{2} - x - \frac{1 \cdot x^3}{2 \cdot 3} - \cdots - \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots 2n} \frac{x^{2n+1}}{(2n+1)} + o(x^{2n+2})$$

(17) 
$$\arcsin(x) = x + \frac{1 \cdot x^3}{2 \cdot 3} + \cdots + \frac{(2n)!}{2^{2n}(n!)^2} \frac{x^{2n+1}}{(2n+1)} + o(x^{2n+2})$$

(18) 
$$\operatorname{argch} \text{ est d\u00e9fini sur } [1, +\infty[ \text{ mais n'admet pas DL au voisinage de } 1^{(5)}$$
  
(tangente verticale).

(19) 
$$\operatorname{argsh}(x) = x - \frac{1 \cdot x^3}{2 \cdot 3} + \cdots + (-1)^n \frac{(2n)!}{2^{2n}(n!)^2} \frac{x^{2n+1}}{(2n+1)} + o(x^{2n+2})$$

Formules de trigonom\u00e9trie

|     | 0 | $\frac{\pi}{6}$      | $\frac{\pi}{4}$      | $\frac{\pi}{3}$      | $\frac{\pi}{2}$ | $\pi$ |
|-----|---|----------------------|----------------------|----------------------|-----------------|-------|
| sin | 0 | $\frac{1}{2}$        | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | 1               | 0     |
| cos | 1 | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$        | 0               | 1     |
| tan | 0 | $\frac{\sqrt{3}}{3}$ | 1                    | $\sqrt{3}$           | non d\u00e9f    | 0     |

$$\begin{aligned} \cos(\frac{\pi}{2} - x) &= \sin(x) & \cos(\frac{\pi}{2} + x) &= -\sin(x) \\ \sin(\frac{\pi}{2} - x) &= \cos(x) & \sin(\frac{\pi}{2} + x) &= \cos(x) \\ \tan(\frac{\pi}{2} - x) &= \frac{1}{\tan(x)} & \tan(\frac{\pi}{2} + x) &= -\frac{1}{\tan(x)} \end{aligned}$$

Transformation des sommes en produits

(1) 
$$\cos(p) + \cos(q) = 2\cos(\frac{p+q}{2})\cos(\frac{p-q}{2})$$

(2) 
$$\cos(p) - \cos(q) = -2\sin(\frac{p+q}{2})\sin(\frac{p-q}{2})$$

(3) 
$$\sin(p) + \sin(q) = 2\sin(\frac{p+q}{2})\cos(\frac{p-q}{2})$$

(4) 
$$\sin(p) - \sin(q) = 2\cos(\frac{p+q}{2})\sin(\frac{p-q}{2})$$

Expressions des sin, cos, tan en fonction de l'angle moiti\u00e9

$$\begin{aligned} t &= \tan\left(\frac{x}{2}\right) & \tan(x) &= \frac{2t}{1-t^2} \\ \cos(x) &= \frac{1-t^2}{1+t^2} & \sin(x) &= \frac{2t}{1+t^2} \end{aligned}$$

(1) 
$$\cos(2a) = 2\cos^2(a) - 1 = 1 - 2\sin^2(a) = \cos^2(a) - \sin^2(a)$$

(2) 
$$\sin(2a) = 2\sin(a)\cos(a) \quad \tan(2a) = \frac{2\tan(a)}{1 - \tan^2(a)}$$

(3) 
$$1 + \cos(x) = 2 \cos^2(\frac{x}{2})$$

(4) 
$$1 - \cos(x) = 2 \sin^2(\frac{x}{2})$$

(5) 
$$1 + \sin(x) = 2 \sin^2(\frac{x}{2} + \frac{\pi}{4})$$

(6) 
$$1 - \sin(x) = 2 \cos^2(\frac{x}{2} + \frac{\pi}{4})$$

R\u00e9sultats classiques d'int\u00e9gration

(1) 
$$\int_{-\infty}^{+\infty} e^{-at^2} \, dt = \sqrt{\frac{\pi}{a}}, \, a > 0 \quad (\text{Int\u00e9grale de Gauss})$$

(2) 
$$\int_0^{+\infty} \frac{\sin t}{t} \, dt = \frac{\pi}{2}, \quad (\text{Int\u00e9grale de Dirichlet})$$

(3) 
$$\Gamma(x) := \int_0^{+\infty} t^{x-1} e^{-t} \, dt = \lim_{n \rightarrow +\infty} \frac{n^x n!}{x(x+1) \cdots (x+n)} \quad x > 0 \quad (\text{Fonction Gamma d'Euler})$$